

Exercise 2.9

Show that

$$\frac{d}{dx} {}_t p_x = {}_t p_x (\mu_x - \mu_{x+t}).$$

Solution

We start with the definition of the survival probability for a life aged x :

$${}_t p_x = \frac{S_0(x+t)}{S_0(x)}.$$

Differentiating with respect to x gives

$$\begin{aligned} \frac{d}{dx} {}_t p_x &= \frac{d}{dx} \left(\frac{S_0(x+t)}{S_0(x)} \right) \\ &= \frac{S'_0(x+t)S_0(x) - S_0(x+t)S'_0(x)}{[S_0(x)]^2}. \end{aligned}$$

Now recall that the force of mortality is

$$\mu_x = -\frac{S'_0(x)}{S_0(x)}.$$

Therefore,

$$S'_0(x) = -\mu_x S_0(x)$$

and similarly,

$$S'_0(x+t) = -\mu_{x+t} S_0(x+t).$$

Substituting these expressions into the derivative, we get

$$\begin{aligned} \frac{d}{dx} {}_t p_x &= \frac{[-\mu_{x+t} S_0(x+t)] S_0(x) - S_0(x+t) [-\mu_x S_0(x)]}{[S_0(x)]^2} \\ &= \frac{-\mu_{x+t} S_0(x+t) S_0(x) + \mu_x S_0(x+t) S_0(x)}{[S_0(x)]^2} \\ &= \frac{S_0(x+t) S_0(x) (\mu_x - \mu_{x+t})}{[S_0(x)]^2} \\ &= \frac{S_0(x+t)}{S_0(x)} (\mu_x - \mu_{x+t}). \end{aligned}$$

Since

$${}_t p_x = \frac{S_0(x+t)}{S_0(x)},$$

it follows that

$$\boxed{\frac{d}{dx} {}_t p_x = {}_t p_x (\mu_x - \mu_{x+t})}.$$